FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XIV

JEE (Main)-2021

TEST DATE: 11-07-2021

ANSWERS, HINTS & SOLUTIONS

Physics

PART – A

SECTION – A

1. A

Sol. In equilibrium

$2T\sin\theta = Mg$

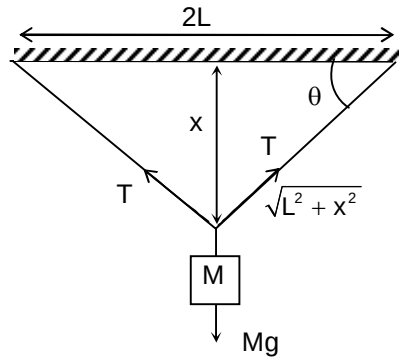
$\Rightarrow \quad T = \dfrac{MgL}{2x}$

$\Delta\ell = 2\left[\sqrt{L^2 + x^2} - L\right] \approx \dfrac{x^2}{L}$

Applying Hooke's Law

$\Rightarrow Y\dfrac{\Delta\ell}{2L} = \dfrac{T}{A}$

$\Rightarrow \left[\dfrac{Mg}{AY}\right]^{1/3} (L)$



2. C

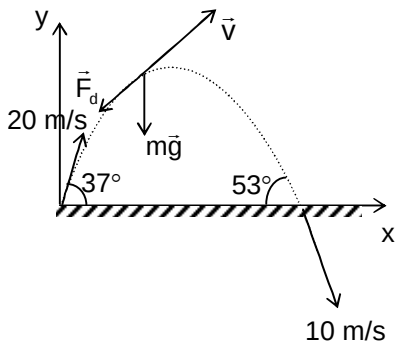
Sol. $m\dfrac{d\vec{v}}{dt} = -k\vec{v} + m\vec{g}$

Along y-axis

$m\dfrac{dv_y}{dt} = -kv_y - mg$

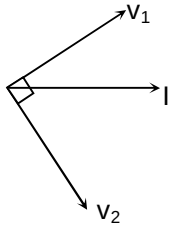
$m\int_{12}^{-8} dv_y = -k\int_0^T v_y dt - mg\int_0^T dt \qquad \left(k\int_0^T v_y dt = 0\right)$

T = 2 sec



3. C

Sol. The only possible case (when resistive component are non-zero and different) has the phasor diagram as shown. Therefore one is capacitive and other is inductive.



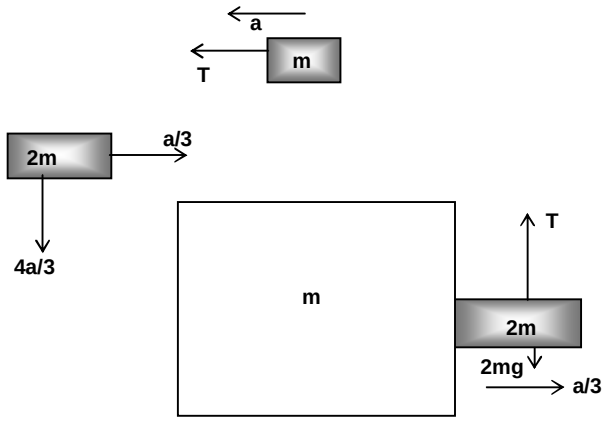
4. D

Sol. T = ma

$$2mg - T = 2m\left(\frac{4a}{3}\right)$$

$$\Rightarrow \quad a = \frac{6g}{11}$$

$$a_B = \sqrt{17}\,\frac{a}{3} = \frac{2\sqrt{17}g}{11}$$

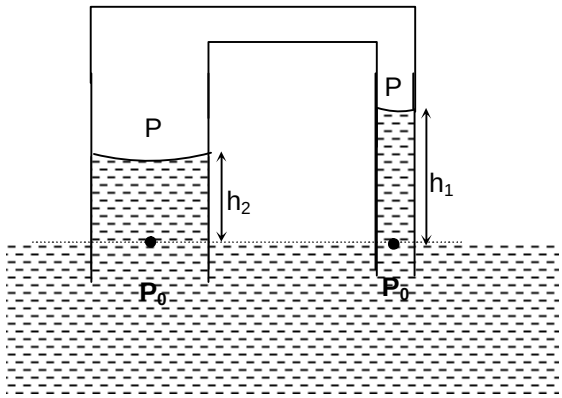


5. B

Sol. $$P - \frac{2T}{3r} + \rho g h_2 = P_0 \ldots(i)$$

$$P - \frac{2T}{r} + \rho g h_1 = P_0 \ldots(ii)$$

From (i) and (ii) $h_1 - h_2 = \frac{4T}{3\rho r g}$



6. D

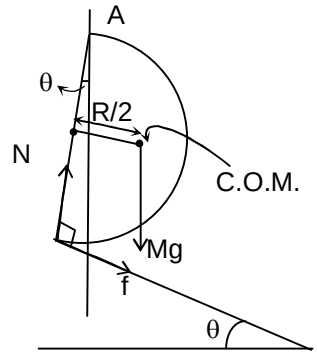
Sol. $\vec{\tau}_A = 0$ (on hemisphere)

$$f2R = Mg\left[\frac{R}{2}\cos\theta - R\sin\theta\right]$$

$$f = \mu N = \tan\theta mg\cos\theta = mg\sin\theta$$

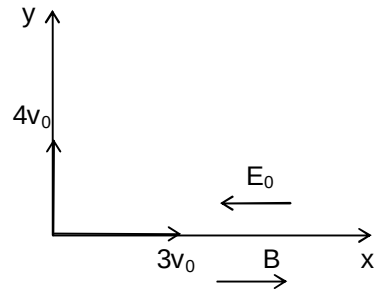
$$mg\sin 4 = Mg\left[\cos\theta - 2\sin\theta\right]$$

$$\tan\theta = \frac{M}{2(2m+M)} = \mu$$



7. B

Sol. Particle goes in helical path.



$$S_{T_n} = 0\ ,\ 3v_0T_n - \frac{1}{2}\left(\frac{qE_0}{m}\right)T_n^2$$

$$T_n = \frac{6mv_0}{qE_0}$$

$$n\left(\frac{2\pi m}{Bq}\right) = \frac{6mv_0}{qE_0} \Rightarrow B = \frac{n\pi E_0}{3v_0}$$

$$B_{min} = \frac{\pi E_0}{3v_0}$$

8. B

Sol. Path difference at A $\Rightarrow S_2A - S_1A = 2\lambda$

Path difference at $\infty \Rightarrow 6\lambda \sin 37° = 3.6\lambda$

Hence the number of minima observed = 10

9. D

Sol. $\frac{2kP_1}{r_1^3} = \frac{2kP_2}{r_2^3}$, where $r_1 + r_2 = \ell$

$$\Rightarrow r_1^3 = \frac{P_1\ell^3}{\left[(P_1)^{1/3} + (P_2)^{1/3}\right]^3},\quad r_2^3 = \frac{P_2\ell^3}{\left[(P_1)^{1/3} + (P_2)^{1/3}\right]^3}$$

When slightly displaced

$$F_{net} = -mx'' = \frac{2kP_2q}{(r_2 - x)^3} - \frac{2kP_1q}{(r_1 + x)^3}$$

$$x'' = -\omega^2 x$$

$$\omega^2 \propto \frac{1}{\ell^4} \Rightarrow T \propto \ell^2 \text{ also } \omega^2 \propto \frac{q}{\varepsilon_0 m}$$

10. A

Sol. Excitation potential of He^+ ion for n = 2 is E.P.

$$\text{E.P.} = \frac{E_2 - E_1}{e} = z^2(13.6\text{V})\left(1 - \frac{1}{(2)^2}\right) = (13.6)3 \text{ V}$$

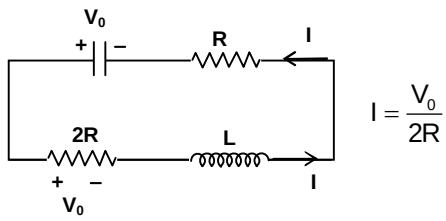
Ionization potential of Li^{2+} ion for n = 4 is I.P.

$$\text{I.P.} = -\frac{E_4}{e} = \left(\frac{3}{4}\right)^2 (13.6\text{V}) = \frac{3}{16}(\text{E.P.})$$

$$\frac{\text{E.P.}}{\text{I.P.}} = \frac{16}{3}$$

11. A

Sol. Just after the switch S is opened, current in the inductor remains same.



Applying KVL in the loop shown.

$$V_0 - L\frac{dI}{dt} - IR - V_0 = 0 \qquad \left(I = \frac{V_0}{2R}\right)$$

$$\frac{dI}{dt} = \frac{-V_0R}{2RL} = \frac{-V_0}{2L}$$

12. D

Sol. $e\sigma A(T^4 - T_0^4) = 280$

$\sigma A(T^4 - T_0^4) = 350$

$e = 0.8$

13. C

Sol. $E = mgh + \frac{1}{2}mv^2$

$P = \frac{dE}{dt} = \left(gh + \frac{v^2}{2}\right)\frac{dm}{dt} = \left(gh + \frac{v^2}{2}\right)\rho\frac{dV}{dt}$

$P = \left(\rho gh + \rho\frac{v^2}{2}\right)\frac{dV}{dt}$, Also $\frac{dV}{dt} = Av$

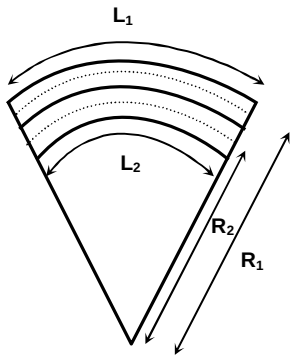
$P = \left[\rho gh + \frac{\rho}{2A^2}\left(\frac{dV}{dt}\right)^2\right]\frac{dV}{dt} = 15$ kW

14. A

Sol. $\frac{L_1}{R_1} = \frac{L_2}{R_2} \Rightarrow \frac{L_1 + L_2}{L_1 - L_2} = \frac{R_1 + R_2}{R_1 - R_2}$

$\frac{2R}{\left(\frac{x}{2}\right)} = \frac{2L + 3L\alpha\Delta\theta}{L\alpha\Delta\theta}$

$R = \frac{x}{4}\left[\frac{2 + 3\alpha\Delta\theta}{\alpha\Delta\theta}\right]$



15. B

Sol. $\left(\frac{\varepsilon}{r+9}\right)^2 9 = \left(\frac{\varepsilon}{r+4}\right)^2 4 = 1$

$\Rightarrow \varepsilon = 5V$

$r = 6\Omega$

16. B

Sol. $dQ = dU + dW \Rightarrow dQ = 3dU$

$C = 3C_v$, $C = 3\left(\frac{5R}{2}\right)$, $C = 7.5\ R$

Heat capacity $= nC = 2C = 15R$

17. D

Sol. At any time t, $\frac{kq}{a} + \frac{dq}{dt}(R) = 0$, $\left(R = \int_a^\infty \frac{1}{\sigma}\frac{dr}{4\pi r^2} = \frac{1}{\sigma 4\pi a}\right)$

$\Rightarrow \int_{Q_0}^{q}\frac{dq}{q} = -\frac{k}{aR}\int_0^t dt$, $q = Q_0 e^{-\frac{\sigma t}{\varepsilon}}$

18. A

Sol. $I_E = I_B + I_C$

$\Delta I_E = \Delta I_B + \Delta I_C$

$8mA = \Delta I_B + 7.9mA \Rightarrow \Delta I_B = 0.1\ mA$

$$\beta = \frac{I_c}{I_B} = \frac{\Delta I_C}{\Delta I_B} = \frac{7.9\ mA}{0.1\,mA} = 79$$

$$\alpha = \frac{I_c}{I_E} = \frac{\Delta I_C}{\Delta I_E} = \frac{7.9\ mA}{8\ mA} = 0.9875 \approx 0.99$$

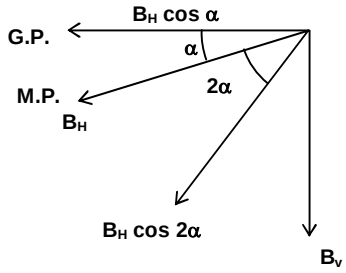
19. B

Sol. $\cot\alpha' = \frac{B_H \cos\alpha}{B_V}$ …(i) $\alpha' = 30°$

$\cot\alpha'' = \frac{B_H \cos 2\alpha}{B_V}$ …(ii) $\alpha'' = 45°$

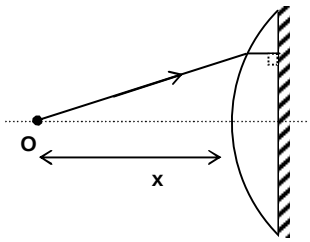
Solving (i) and (ii) we get declination

$\alpha = 30°$



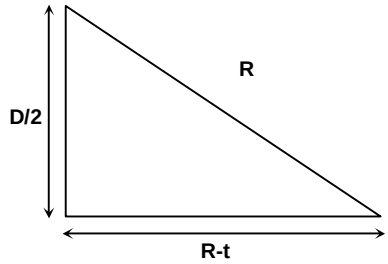
20. C

Sol. Ray diagram so that image coincides with the object



$$R^2 = (R-t)^2 + \frac{D^2}{4}$$

$D^2 \approx 8Rt$



$$\frac{\mu}{\infty} - \frac{1}{-x} = \frac{\mu - 1}{R}$$

$$x = \frac{R}{\mu - 1} = \frac{D^2}{8t(\mu - 1)}$$

Distance of object from plane surface = x + t

$$= \frac{D^2}{8t(\mu-1)} + t = \frac{D^2 + 8(\mu-1)t^2}{8t(\mu-1)}$$

SECTION – B

21. 00001.00

Sol. Applying work energy theorem on block m_1

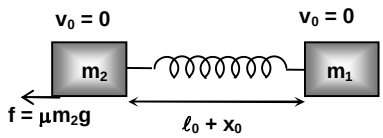
$$-\mu m_1 g x_0 - \frac{1}{2}kx_0^2 = -\frac{1}{2}m_1 v_0^2 \quad \text{…(i)}$$

$$\mu m_2 g = kx_0 \quad \text{…(ii)}$$

Therefore solving (i) and (ii) we get

$$v_0 = \mu g\sqrt{\frac{m_2}{km_1}(2m_1 + m_2)}$$

$v_0 = 1 m/s$



22. 00002.00

Sol. $(2n+1)\frac{\lambda}{4} = 53.8 \text{ cm} + e$ (e = end correction)

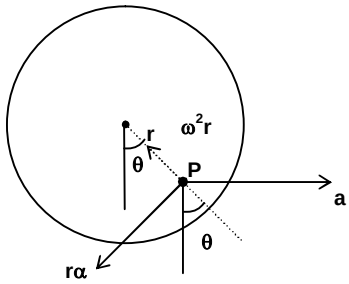
$(2n+3)\frac{\lambda}{4} = 75.8 \text{ cm} + e$

Solving these we get n = 2, e = 1.2 cm and λ = 44 cm

$e = 0.6\, r \Rightarrow r = 2$ cm

23. 00000.75

Sol. These is only one point on the disc with zero acceleration in case of pure rolling.



$\vec{a}_P = 0$

$\frac{a}{\sin 90°} = \frac{r\alpha}{\cos\theta} = \frac{\omega^2 r}{\sin\theta}$

$\alpha = \frac{a}{R}$, r = 4m, R = 5m, a = 5 m/s^2

$\Rightarrow \cos\theta = 4/5$

$\omega^2 = \frac{a\sin\theta}{r} = 5\left(\frac{3}{5}\right)\frac{1}{4}$

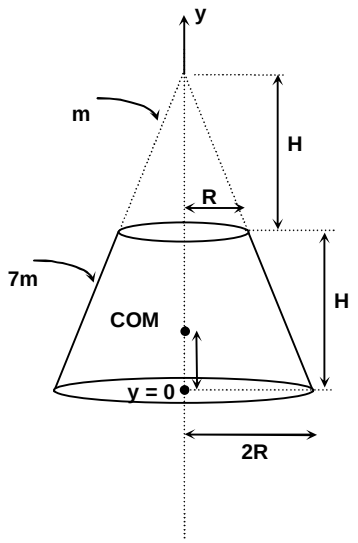
$\omega^2 = \frac{3}{4} \Rightarrow k = 0.75$

24. 00005.50

Sol. $y_{cm} = \frac{2H}{4}$ (for complete system)

$7mh + m\left(H + \frac{H}{4}\right) = 8m\left(\frac{2H}{4}\right)$

$\frac{h}{H} = \frac{11}{28} \Rightarrow \frac{14h}{H} = 5.50$



25. 00000.99

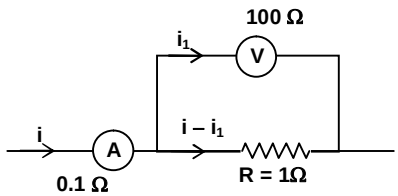
Sol. $i_1 100 = 1(i - i_1)$

$i_1 = \frac{i}{101}$

$v_{measured} = i_1(100) = \frac{100i}{101}$

$I_{measured} = i$

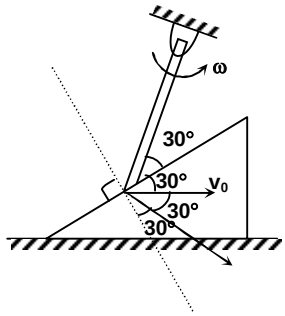
$R_{measured} = \frac{V_m}{I_m} = \frac{100}{101}$



$$\%R = \frac{R - R_{meas}}{R} \times 100 = \frac{\left(1 - \frac{100}{101}\right) \times 100}{1} = \frac{100}{101}\% = 0.99$$

26. 00003.00

Sol. Components of velocities of point of contact normal to the inclined plane is same.



$v_0 \cos 60° = \omega L \cos 30°$

$$\left(\frac{v_0}{\omega L}\right)^2 = 3$$

27. 00075.00

Sol. $v = \frac{k}{x}$, $100 = \frac{k}{2}$, k = 200

$$\int_{x_1}^{x_2} x\,dx = k\int_0^T dt$$

$$\frac{x_2^2 - x_1^2}{2} = kT$$

T = 75 sec

28. 00004.14

Sol. $hf - \phi = eV_s$

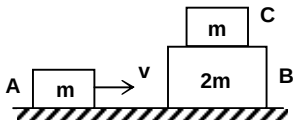
$$V_s = \left(\frac{h}{e}\right) f - \frac{\phi}{e}$$

$$\frac{h}{e} = \text{slope} = \frac{1.656}{4 \times 10^{14}} \text{ V-s}$$

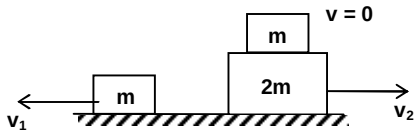
$$\frac{h}{e} = 4.14 \times 10^{-15} \text{ V-s}$$

29. 00002.25

Sol.



Before collision



Just after collision

Conservation of momentum

$\Rightarrow mv = -mv_1 + 2mv_2$

$2v_2 - v_1 = v$ …(i)

$v_2 + v_1 = v$ $(e = 1)$ …(ii)

From (i) and (ii) $v_2 = \frac{2v}{3}$

Finally common velocity of B and C is

$$\frac{2v}{3}\frac{(2m)}{(2m+m)} = \frac{4v}{9} = \frac{v}{2.25}$$

30. 00000.75

Sol. In equilibrium $mg = f_B$

$$2\rho\frac{4}{3}\pi\left(R_2^3 - R_1^3\right)g = \rho\frac{2}{3}\pi R_2^3 g$$

$$\Rightarrow \left(\frac{R_1}{R_2}\right)^3 = \frac{3}{4} = 0.75$$

Chemistry

PART – B

SECTION – A

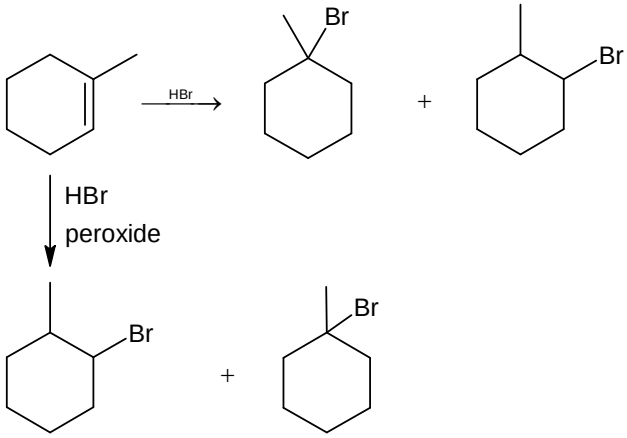
31. B

Sol. Bond energy is inversely related to bond length for diatomic molecules.

32. B

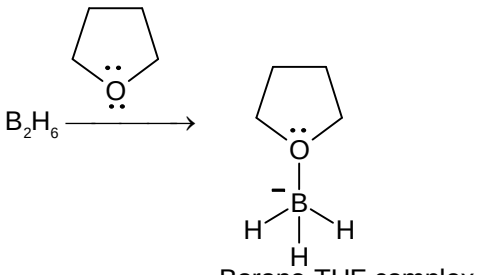
33. D

Sol.



34. B

Sol.

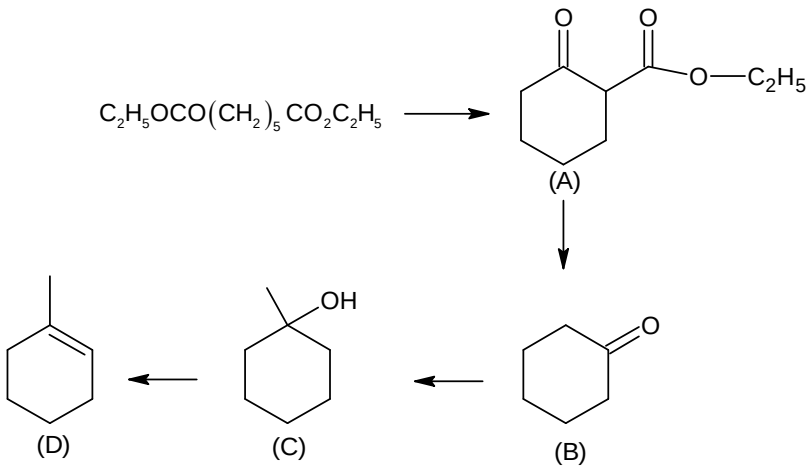


Borane-THF complex

35. C

36. D

Sol.

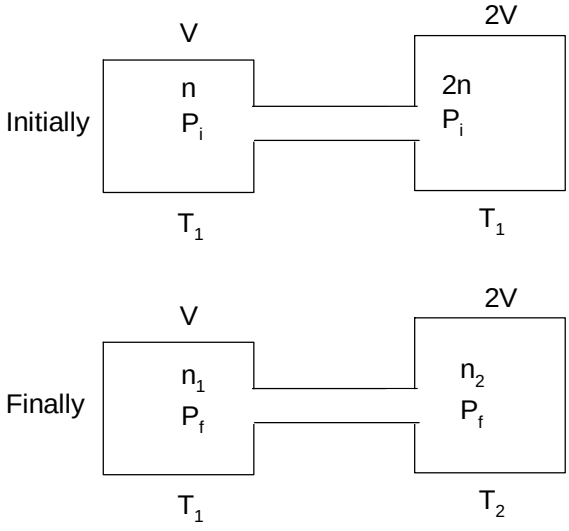


37. D

Sol. The possible acids are $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-H$, $CH_3-CH_2-\overset{\overset{^{18}O}{\|}}{C}-O-H$ and $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-\underset{18}{O}-H$. The possible alcohol is CH_3-CH_2-O-H.

38. C

Sol.



$3n = n_1 + n_2$

$$\frac{3VP_i}{RT_1} = \frac{P_f V}{RT_1} + \frac{P_f 2V}{RT_2}$$

$$\Rightarrow \frac{3P_i}{T_1} = P_f\left(\frac{1}{T_1} + \frac{2}{T_2}\right)$$

$$\Rightarrow P_f = \frac{3P_i}{T_1}\left(\frac{T_1 \times T_2}{T_2 + 2T_1}\right)$$

$$\Rightarrow P_f = 3P_i\left(\frac{T_2}{T_2 + 2T_1}\right)$$

39. D

Sol. For uncharged acid

$HA \rightleftharpoons H^+ + A^-$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

For charged acid

$HA^+ \rightleftharpoons H^+ + A$

$$K_a = \frac{[H^+][A]}{[HA^+]}$$

Polar solvent has greater stability effect on charged species. Hence, in case of uncharged acid like carboxylic acid, alcohol and phenol, it stabilizes H^+ and A^- more than it stabilizes HA.

40. C

Sol. Formation of 3 membered cyclic ether is more favoured over 4 membered. 5- or 6-membered cyclic ether rings are more stable. 7-membered and higher rings are too unstable.

41. D

42. B

Sol. I and III may yield secondary and tertiary amine as well.

$$CH_3-CH_2-CH_2-Br \xrightarrow{NaN_3} CH_3-CH_2-CH_2-N_3 \xrightarrow{H_2,\,Pd/C} CH_3-CH_2-CH_2-NH_2$$

43. D

44. B

45. B

46. B

47. B

Sol. Fehling test is given by aldehyde, hemiacetal; even formic acid also gives the test. Ketone, acetal don't give this test.

48. B

49. C

Sol. Check NCERT for nitrogen oxides.

50. C

Sol. H_2O_2 oxides SO_2 to SO_4^{2-} then $BaCl_2$ form precipitate with that. That is not going to happen with CO_2.

SECTION – B

51. 00050.00

Sol. After electrolysis (knowing the percentage yield 50%)

Weight of aluminium deposited, $W = \frac{50}{100} \times Z \times Q$

$$W = \frac{1}{2} \times \frac{E}{F} \times 2F$$

$$= \frac{1}{2} \times \frac{27}{3F} \times 2F = \frac{27}{3} g$$

$$4Al + 3O_2 \longrightarrow 2Al_2O_3$$

Amount of aluminium oxide would be formed (assuming 100% yield) = 17 g
But it was formed = 8.5 g
Required percentage yield = 50%.

52. 00001.00

Sol. The concentration difference between the body fluid of blood cells and the solution in which it is dissolved should be such that the osmotic pressure difference should be less than the withstanding pressure, i.e. 0.4205 atm.

$\pi = (C_2 - C_1)RT$

53. 00025.00

Sol. Overall complex dissociation constant=$\frac{1}{\text{overall stability constant}}$

54. 00028.72

Sol. During X to Y, work done = 0

The temperature and volume at Y are 750 K and 24.6 Lt, respectively.

During Y to Z

$$W = -nRT\ell n\frac{V_2}{V_1}$$

$$= -2\times 8.314\times 750\times 2.303\log\frac{V_2}{V_1}$$

$$= -28720.71\log\frac{2.46}{24.6}$$

= 28720.71 Joules

= 28.72 kJ

55. 00000.50

Sol. Degree of dissociation of N_2O_4 is 0.5.

56. 20564.44

Sol. $\bar{\nu} = \bar{\nu}_1 + \bar{\nu}_2$

$$R_\infty\left(\frac{1}{2^2}-\frac{1}{5^2}\right) = R_\infty\left(\frac{1}{4^2}-\frac{1}{5^2}\right)+\bar{\nu}_2$$

Find $\bar{\nu}_2$.

57. 00002.00

Sol. $$\log\frac{x}{m} = \underbrace{\log K}_{2.85} + \frac{1}{n}\log p$$

$$\text{Slope} = \frac{1}{n} = \frac{1}{4}$$

$$\log\frac{x}{0.002} = 2.85 + \frac{1}{4}\log 4$$

$$\Rightarrow \log\frac{x}{0.002} = 2.85 + 0.15$$

$$\Rightarrow \log\frac{x}{0.002} = 3$$

$$x = 2.00$$

58. 00004.00

Sol. $$\left[CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)\right]$$

1	0	0
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$

$$K_P = P_{eq} = \frac{nRT}{V}$$

$= \frac{1}{2} \times \frac{0.082 \times 1600}{16.4}$

$= \frac{1600}{4} \times 10^{-2}$

= 4 atm.

59. 00209.00

Sol. As, 4 formula unit of AB are present per unit cell. After the replacement, the molecular formula unit present per unit cell will be A_3B_4C.

Required mass $= 3 \times 24 + 32 \times 4 + 9$

60. 00001.00

Sol. $r = k[\text{sucrose}]^1[H^+]^b$ because half-life remains constant with change in concentration of sugar; a = 1.

At any particular pH

$r = K[\text{sucrose}], k[H^+]^b = K$

$\Rightarrow \frac{0.69}{t_{1/2}} = k[H^+]^b = K$ ... (1)

$\Rightarrow \frac{0.69}{k} = t_{1/2}[H^+]^b = \text{constant}$

$50\left(10^{-4}\right)^b = 500\left(10^{-5}\right)^b$

= 1.00

Mathematics

PART – C

SECTION – A

61. B

Sol. From $\begin{cases} x > 0, \\ x \neq 1, \\ 2x^2 + x - 1 > 0, \end{cases}$ we get, $x > \frac{1}{2}, x \neq 1$

Furthermore, $\log_x (2x^2 + x - 1) > \log_x 2 - 1$

$\Rightarrow \log_x (2x^3 + x^2 - x) > \log_x 2$

$\Rightarrow \begin{cases} 0 < x < 1, \\ 2x^3 + x^2 - x < 2, \end{cases}$ or $\begin{cases} x > 1, \\ 2x^3 + x^2 - x > 2 \end{cases}$

Then we have $x > \frac{1}{2}$ and $x \neq 1$

62. C

Sol. Since $5x - a \leq 0 \Rightarrow x \leq \frac{a}{5}$, $5x - b > 0 \Rightarrow x > \frac{b}{6}$

In order to satisfy $A \cap B \cap N = \{2, 3, 4\}$, we have $\begin{cases} 1 \leq \frac{b}{6} < 2, \\ 4 \leq \frac{a}{5} < 5, \end{cases}$ or $\begin{cases} 6 \leq b < 12, \\ 20 \leq a < 25 \end{cases}$

So the number of pairs (a, b) is $\binom{6}{1} \cdot \binom{5}{1} = 30$

63. D

Sol. Since each has equally 9 different possible results for A and B to draw a ball from the packet independently, the total number of possible events is $9^2 - 81$. From $a - 2b + 10 > 0$ we get $2b < a + 10$. We find the when b – 1, 2, 3, 4, 5, a can take any value in 1, 2, ..., 9 to make the inequality hold. Then we have $9 \times 5 = 45$ admissible events.
When b – 6, a can be 3, 4, ..., 9 and there are 7 admissible events.
When b = 8, a can be 7, 8, 9 and there are 3 admissible events.
When b – 9, a can only be 9 and there is 1 admissible event.
So the required probability is $\frac{45 + 7 + 5 + 3 + 1}{81} = \frac{61}{81}$

64. A

Sol. Let $x = \frac{2}{3}a$, then we have $|a| \leq \frac{1}{3}$. Therefore (B) and (D) are excluded. By symmetry, (C) is also excluded. Then only (A) can be correct.

65. C

Sol. Let $f(x) = |2x + 2|x| + 2\alpha| + |2x - 2|x| - 2\alpha|$
Clearly f(x) is an even function f(x) = 2016 will have 3 solution if f(0) = 2016
$\Rightarrow \lambda = \pm 504$ for $\alpha = 504$, then equation has only one solution

66. C

Sol. As function is onto

Hence, $0 \leq \log\left(\sqrt{9x^2 - 12x + \lambda + 1}\right) < \infty$

$\Rightarrow \sqrt{9x^2 - 12x + \lambda} \geq 0$

$\therefore\ 9x^2 - 12x + \lambda \geq 0$ for this D = 0

$\Rightarrow 12^2 - 4 \times 9\lambda = 0\ \therefore\ \lambda = 4$

67. D

Sol. Put $1 - x^{2021} = t^2$

68. D

Sol. Put $\frac{3}{x} \to y$

The limit converts into $\lim\limits_{y\to\infty}\left(\frac{y+1}{y-1}\right)^{\frac{y}{3}} = e^{\frac{2}{3}}$

69. D

Sol. Let x = tan θ ; $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$ when $\theta \neq \frac{\pi}{4} \Rightarrow f(x) = \frac{1}{\sqrt{2}\sin\left(\theta - \frac{\pi}{4}\right)}$

70. A

Sol. $f(x) = \begin{cases} x^2 + 2 & 1 \leq x < 2 \\ \frac{x^2+2}{2} & 2 \leq x < 3 \end{cases}$ clearly $f(2^-) = 6$, $f(2^+) = 3$

So, f(x) is not increasing in [1, 3]

71. D

Sol. $\bar{a} = \frac{\hat{i} - \hat{j}}{\sqrt{2}}$

72. B

Sol. Required probability = ${}^3C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 \times \frac{1}{4}^3 + {}^4C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^2 \cdot \frac{1}{4} + {}^5C_1 \cdot \frac{1}{4} \cdot \left(\frac{3}{4}\right)^4 \cdot \frac{1}{4} ..... = \frac{27}{32}$

73. C

Sol. Clearly λ is negative

So, $\tan^{-1}\lambda + \tan^{-1}\frac{1}{\lambda} = \tan^{-1}\lambda - \cot^{-1}(-\lambda)$

$\Rightarrow -\left[\tan^{-1}(-\lambda) + \cot^{-1}(-\lambda)\right] = -\frac{\pi}{2}$

74. B

Sol. $a^2 + 4b = 2a(b - 3)$ a, b $\in$ I

Clearly for increasing AP a = 4, and b = 10 in the only possible integral value

75. D

Sol. $\frac{\left|z + \frac{3}{2}\right|}{\left|z + \frac{4}{3}\right|} = \frac{3}{2}$ appolorious form of circle

$\left(-\frac{7}{5}, 0\right)$ and (–1, 0) end points of diameter

Centre $\left(-\frac{6}{5}, 0\right)$ and radius $\frac{1}{5}$

76. C

Sol. $n_1 = 35,\ \bar{x}_1 = 80,\ \sigma_1 = 4$

$n_1 = 65,\ \bar{x}_2 = 60,\ \sigma_2 = 4$

$\sigma^2 = \frac{n_1\left(\sigma_1^2 + d_1^2\right) + n_2\left(\sigma_2^2 + d_1^2\right)}{n_1 + n_2}$ combined mean = $\frac{35\times 80 + 65\times 60}{100} = 67$

$\therefore\ d_1 = 80 - 67 = 13,\ d_2 = 60 - 67 = -7\ \therefore\ \sigma^2 = \frac{35(16+169) + 65(9+49)}{100}$

$\sigma^2 \Rightarrow 102.45\ \therefore\ \sigma = \sqrt{102.45} = 10.12$

77. B

Sol. R and S be two equivalence relation on a set A

$\Rightarrow R \cap S$ is an equivalence relation on A

78. C

Sol. Given table

p	q	$\sim p \vee q$	$\sim p \wedge \sim q$	$(\sim p \vee q) \wedge (\sim p \wedge \sim Q)$
T	T	T	F	F
T	F	F	F	F
F	T	T	F	F
F	F	T	T	T

79. A

Sol. By property $PQ \cdot PQ' = b^2$ where b is the length of semi-conjugate axis

80. C

Sol. Director Circle is directrix of parabola which is x = –11

Radius r = 12

SECTION – B

81. 00012.00

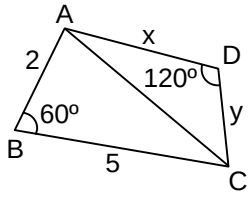
Sol. Area of ABC + Area of ADC

$= 4\sqrt{3} = \frac{5\sqrt{3}}{2} + \frac{\sqrt{3}}{4}xy$

$\Rightarrow xy = 6$

And equating length of AC in ΔABC and ΔADC, we get

$x^2 + y^2 + xy = 19$, we get x = 2, y = 3 or x = 3, y = 2



82. 02012.00

Sol. $\lim_{x\to 0} \frac{\int_{-x}^{x}\left(t\cdot\sin(2010t) + 2009t + 1006\right)dt}{x} = \frac{2012}{1}$ (By L Hospital Rule)

83. 00002.00

Sol. $\dfrac{-\dfrac{dx}{x^2}+\dfrac{dy}{y^2}}{\dfrac{(x-y)}{xy}}+y\,dy=0 \Rightarrow \dfrac{d\left(\dfrac{1}{y}-\dfrac{1}{x}\right)}{\left(\dfrac{1}{y}-\dfrac{1}{x}\right)}=ydy \Rightarrow \log\dfrac{(x-y)}{xy}=\dfrac{y^2}{2}+c$

So, k = 2

84. 00010.00

Sol. $M^2=\begin{bmatrix}a & -360\\ b & c\end{bmatrix}\begin{bmatrix}a & -360\\ b & c\end{bmatrix}=\begin{bmatrix}a^2-360b & -360(a+c)\\ b(a+c) & -360b+c^2\end{bmatrix}$

Which states that a = –c and a^2 – 360b = 0 for 360b to be perfect square the minimum value of b is 10

85. 00063.00

Sol. Required ways = ${}^6C_1+{}^6C_2+{}^6C_3+{}^6C_4+{}^6C_5+{}^6C_6=63$

86. 00210.00

Sol. Differentiate the equation twice w.r.t x and put x = 0, we get a_2 = 210

87. 02047.00

Sol. $(AB)^2 = ABAB = A^3B^2$

$(AB)^3 = AB\cdot A^3B^2 = A^7\cdot B^3$

$(AB)^4 = A^{15}\cdot B^4$

$(AB)^n = A^{(2^n-1)}\cdot B^n$

88. 01260.00

Sol. $N=\dfrac{9!}{4!\,3!\,2!}=1260$

89. 00008.00

Sol. $(y+1)^2 = 4(x-1)$

Equation of normal passing in slope form $(y+1) = m(x-1) - 2m - m^2$

Let P(h, k) be the point from where the normal passes

$\Rightarrow m^3 - m(h-3) + k + 1 = 0$; $m_1m_2m_3 = -(k+1) \Rightarrow m_3 = k+1$

$\Rightarrow (k+1)^3 - (k+1)(h-3) + (k+1) = 0 \Rightarrow (y+1)^2 - (x-4) = 0$

Comparing k = –2, $\lambda = 4$; $\lambda - 2k = 8$

90. 00256.00

Sol. $N = D_6 - D_4 = 265 - 9 = 256$